```python

# Q1 ans-

class Person:
    def __init__(self, name, id):
        self.name = name
        self.id = id

    def Display(self):
        print(f"{self.name}, {self.id}")

class Emp(Person):
    def __init__(self, name, id):
        super().__init__(name, id)

    def Print(self):
        print("Emp class called")

emp_details = Emp("Mayank", 103)
emp_details.Display()
print("fix using super() to call parent class __init__")
print()


```

Problems 7 | Output | Debug Console | Terminal | Ports | Spell Checker 7

```
assignment_1
assignment_2,3
assignment_4
assignment_5
PS C:\Users\vttha\OneDrive\Desktop\sem 3\python programming\assignments>
 *  History restored

PS C:\Users\vttha\OneDrive\Desktop\sem 3\python programming\assignments> cd '.\6_Practical 6 - Classes and Objects\'
PS C:\Users\vttha\OneDrive\Desktop\sem 3\python programming\assignments\6_Practical 6 - Classes and Objects> python .\6_practical.py
Mayank, 103
fix using super() to call parent class __init__
```

```python
# Q2 ans-

def is_anagram(str1, str2):
    str1 = str1.replace(" ", "").lower()
    str2 = str2.replace(" ", "").lower()

    return sorted(str1) == sorted(str2)

str1 = "listen"
str2 = "silent"

if is_anagram(str1, str2):
    print(f"'{str1}' and '{str2}' are anagrams")
else:
    print(f"'{str1}' and '{str2}' are not anagrams")

str3 = "hello"
str4 = "world"

if is_anagram(str3, str4):
    print(f"'{str3}' and '{str4}' are anagrams")
else:
    print(f"'{str3}' and '{str4}' are not anagrams")

print()
```

Problems 7 | Output | Debug Console | Terminal | Ports | Spell Checker 7

powershell - 6_Practical 6 - Classes and Objects

```
PS C:\Users\vttha\OneDrive\Desktop\sem 3\python programming\assignments\6_Practical 6 - Classes and Objects> python .\6_practical.py
'listen' and 'silent' are anagrams
'hello' and 'world' are not anagrams
```

```python

# Q3 ans-

def is_armstrong(num):
    num_str = str(num)
    num_digits = len(num_str)
    sum_of_powers = sum(int(digit) ** num_digits for digit in num_str)

    return sum_of_powers == num

test_numbers = [153, 370, 9474, 123, 1634, 8208]

for num in test_numbers:
    if is_armstrong(num):
        print(f"{num} is an Armstrong number")
    else:
        print(f"{num} is not an Armstrong number")

print()

```

Problems 7 | Output | Debug Console | Terminal | Ports | Spell Checker 7 | powershell - 6_Practical 6 - Classes and O

```
PS C:\Users\vttha\OneDrive\Desktop\sem 3\python programming\assignments\6_Practical 6 - Classes and Objects> python .\6_practical.py

153 is an Armstrong number
370 is an Armstrong number
9474 is an Armstrong number
123 is not an Armstrong number
1634 is an Armstrong number
8208 is an Armstrong number
```

```python
# Q4 ans-
class Vehicle:
    color = "white"

    def __init__(self, max_speed, mileage):
        self.max_speed = max_speed
        self.mileage = mileage
        self.seating_capacity = 0

    def assign_seating_capacity(self, capacity):
        self.seating_capacity = capacity

    def display(self):
        print(f"Max Speed: {self.max_speed} kmph")
        print(f"Mileage: {self.mileage} kmpl")
        print(f"Color: {self.color}")
        print(f"Seating Capacity: {self.seating_capacity}")

print("\nTask-1,2,3,4: Vehicle class with attributes and methods")
print("-" * 80)

car1 = Vehicle(200, 20)
car1.assign_seating_capacity(5)

car2 = Vehicle(180, 25)
car2.assign_seating_capacity(4)

print("\nCar 1 Details:")
car1.display()

print("\nCar 2 Details:")
car2.display()
print("\nTask-5: Two Vehicle objects created with specified properties")
print("-" * 80)
print("Car 1: Max Speed = 200 kmph, Mileage = 20 kmpl, Seating = 5")
print("Car 2: Max Speed = 180 kmph, Mileage = 25 kmpl, Seating = 4")
```

```
Task-1,2,3,4: Vehicle class with attributes and methods
--------------------------------------------------------------------------------

Car 1 Details:
Max Speed: 200 kmph
Mileage: 20 kmpl
Color: white
Seating Capacity: 5

Car 2 Details:
Max Speed: 180 kmph
Mileage: 25 kmpl
Color: white
Seating Capacity: 4

Task-5: Two Vehicle objects created with specified properties
--------------------------------------------------------------------------------
Car 1: Max Speed = 200 kmph, Mileage = 20 kmpl, Seating = 5
Car 2: Max Speed = 180 kmph, Mileage = 25 kmpl, Seating = 4
PS C:\Users\vttha\OneDrive\Desktop\sem 3\python programming\assignments\6_Practical 6 - Classes and Objects>
```